

```

# Joyal John Mathew 25 MCA028
#1)
x<-c(20,22,23,25,26,27)
y<-c(31,29,32,37,35,34)
correlation <-cor(x,y)
print(correlation)
plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
x<-c(20,22,23,25,26,27)
y<-c(31,29,32,37,35,34)
model<-lm(y~x)
summary(model)
plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
abline(model,col="red",lwd=2)

#2)
x<-c(6,2,10,4,8)
y<-c(9,11,5,8,7)
correlation <-cor(x,y)
print(correlation)
plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
x<-c(6,2,10,4,8)
y<-c(9,11,5,8,7)
model<-lm(y~x)
summary(model)
plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
abline(model,col="red",lwd=2)

#3)
x<-c(9,8,7,6,5,4,3,2,1)
y<-c(15,16,14,13,11,12,10,8,9)
correlation <-cor(x,y)
print(correlation)
plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
x<-c(9,8,7,6,5,4,3,2,1)
y<-c(15,16,14,13,11,12,10,8,9)
model<-lm(y~x)
summary(model)
plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
abline(model,col="red",lwd=2)

#4)
x<-c(77,54,27,52,14,35,90,25,56,60)
y<-c(35,58,60,40,50,40,35,56,34,42)
correlation <-cor(x,y)
print(correlation)
plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
x<-c(77,54,27,52,14,35,90,25,56,60)
y<-c(35,58,60,40,50,40,35,56,34,42)
model<-lm(y~x)
summary(model)
plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
abline(model,col="red",lwd=2)

```

```

> # Joyal John Mathew 25 MCA028
> #1)
> x<-c(20,22,23,25,26,27)
> y<-c(31,29,32,37,35,34)
> correlation <-cor(x,y)
> print(correlation)
[1] 0.7320418
> plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
> x<-c(20,22,23,25,26,27)
> y<-c(31,29,32,37,35,34)
> model<-lm(y~x)
> summary(model)

```

Call:

```
lm(formula = y ~ x)
```

Residuals:

1	2	3	4	5	6
1.0813	-2.5263	-0.3301	3.0622	0.2584	-1.5455

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.8421	8.9599	1.545	0.1973
x	0.8038	0.3740	2.149	0.0981

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.208 on 4 degrees of freedom

Multiple R-squared: 0.5359, Adjusted R-squared: 0.4199

F-statistic: 4.619 on 1 and 4 DF, p-value: 0.09808

```

> plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
> abline(model,col="red",lwd=2)
>
> #2)
> x<-c(6,2,10,4,8)
> y<-c(9,11,5,8,7)
> correlation <-cor(x,y)
> print(correlation)
[1] -0.9192388
> plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
> x<-c(6,2,10,4,8)
> y<-c(9,11,5,8,7)
> model<-lm(y~x)
> summary(model)

```

Call:

```
lm(formula = y ~ x)
```

Residuals:

1	2	3	4	5
1.0	0.4	-0.4	-1.3	0.3

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.9000	1.0661	11.162	0.00154 **
x	-0.6500	0.1607	-4.044	0.02721 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.017 on 3 degrees of freedom

Multiple R-squared: 0.845, Adjusted R-squared: 0.7933

F-statistic: 16.35 on 1 and 3 DF, p-value: 0.02721

```
> plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
> abline(model,col="red",lwd=2)
>
> #3)
> x<-c(9,8,7,6,5,4,3,2,1)
> y<-c(15,16,14,13,11,12,10,8,9)
> correlation <-cor(x,y)
> print(correlation)
[1] 0.95
> plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
> x<-c(9,8,7,6,5,4,3,2,1)
> y<-c(15,16,14,13,11,12,10,8,9)
> model<-lm(y~x)
> summary(model)
```

Call:

lm(formula = y ~ x)

Residuals:

Min	1Q	Median	3Q	Max
-1.15	-0.80	0.05	0.80	1.15

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.2500	0.6641	10.92	1.20e-05 ***
x	0.9500	0.1180	8.05	8.76e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9142 on 7 degrees of freedom

Multiple R-squared: 0.9025, Adjusted R-squared: 0.8886

F-statistic: 64.79 on 1 and 7 DF, p-value: 8.763e-05

```
> plot(x,y,main="Linear Regression", xlab = "x",ylab="Y",pch=19)
> abline(model,col="red",lwd=2)
>
> #4)
> x<-c(77,54,27,52,14,35,90,25,56,60)
> y<-c(35,58,60,40,50,40,35,56,34,42)
> correlation <-cor(x,y)
> print(correlation)
[1] -0.6656882
> plot(x,y,main="Scatter Plot",xlab="x values" ,ylab="y values",pch=19)
> x<-c(77,54,27,52,14,35,90,25,56,60)
```

```
> y<-c(35,58,60,40,50,40,35,56,34,42)
> model<-lm(y~x)
> summary(model)
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.030	-4.675	-1.014	3.570	14.407

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	58.7866	6.0216	9.763	1.02e-05	***
x	-0.2814	0.1115	-2.523	0.0356	*

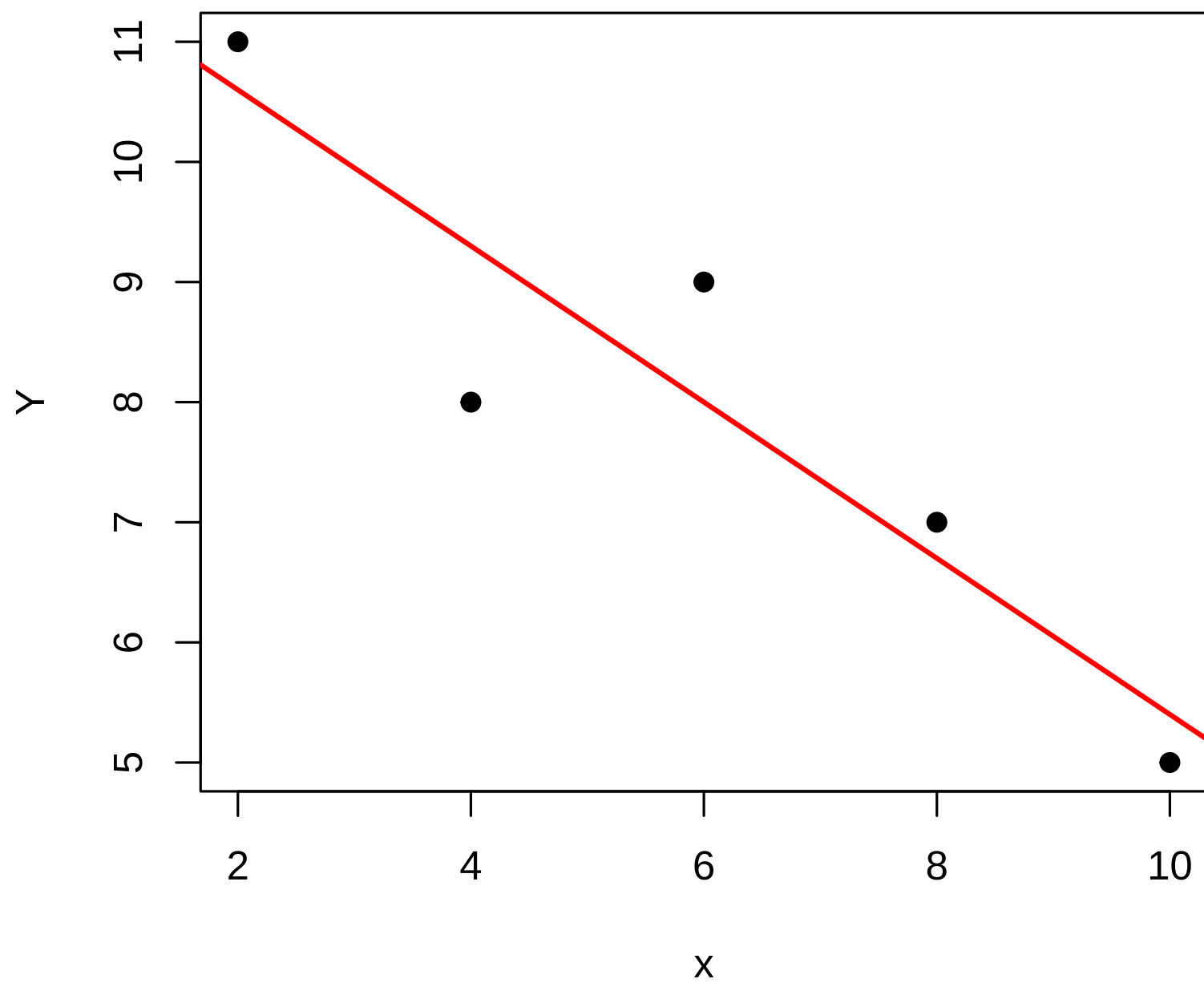
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.002 on 8 degrees of freedom

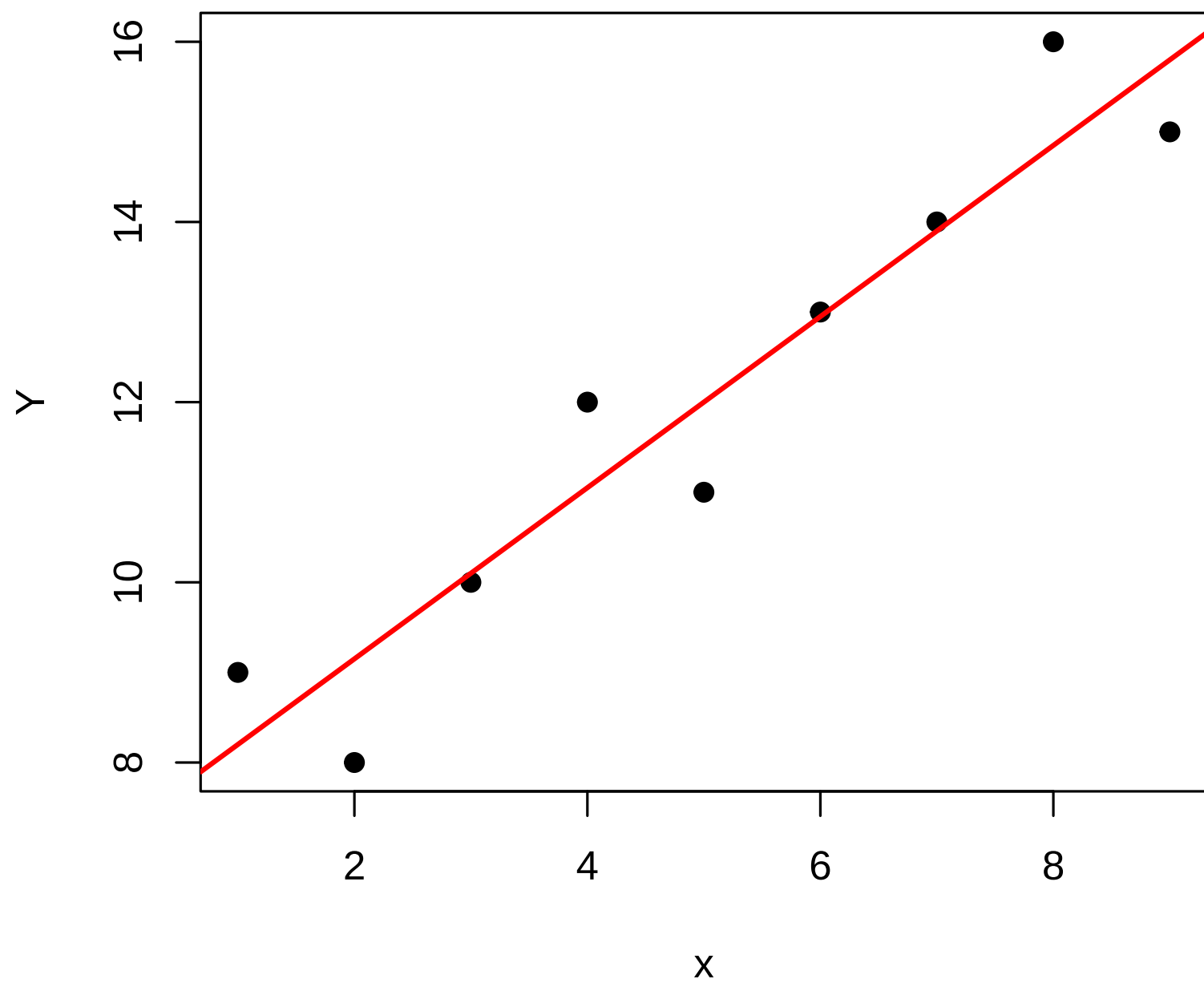
Multiple R-squared: 0.4431, Adjusted R-squared: 0.3735

F-statistic: 6.366 on 1 and 8 DF, p-value: 0.03563

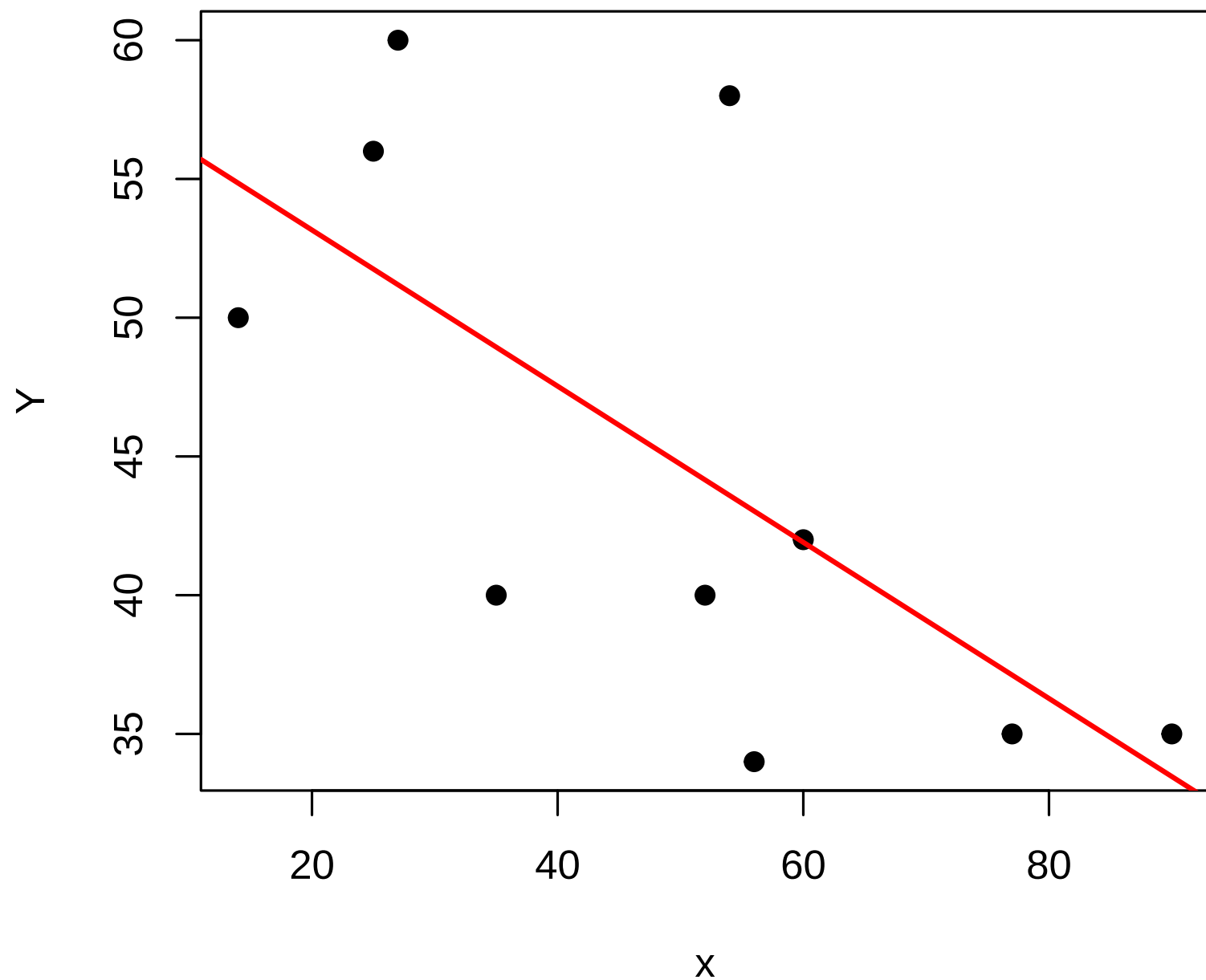
Linear Regression



Linear Regression



Linear Regression



Linear Regression

